

Course 6033A

Semester VI

Theory

4 Credits

Energy Conservation and Audit

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes

Electrical & Electronics Engineering

Contact hours

60

Teaching scheme

3:1:0:0

Credits

4

CIE / ESE

Theory CIE 40 / Theory ESE 60

Self-learning

5.5 h/week

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 6033A as Energy Conservation and Audit in Semester VI for Electrical & Electronics Engineering. The official SITTTTR detailed course page was checked at the indexed link and returned “Requested file not found.” With the user's approved public-source approach, this handbook is transparently based on NPTEL IIT Kharagpur’s Building Energy Systems and Auditing course and polytechnic energy conservation curricula. It does not claim to reproduce official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. Energy audits, equipment specifications, financial models and conservation measures must be based on approved engineering plans, certified hardware data, current safety standards and competent professional review.

Verified source note: Verified sources support energy audit methodology, conservation in electrical/thermal systems, and energy economics. The four-module sequence is a learning path, not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60

Contact hours

4

Credits

Theory CIE 40

CIE

Theory ESE 60

ESE

Course introduction

Energy Conservation and Audit studies the systematic approach to optimizing energy use in industrial, commercial and residential sectors. It develops the ability to analyze energy consumption patterns, conduct preliminary and detailed energy audits, evaluate energy conservation measures in electrical and thermal systems, and perform economic analysis of energy saving projects while respecting the technical and safety boundaries of electrical and industrial engineering.

Malayalam support: ഹാൻഡ്‌ബുക്ക് സിദ്ധാന്തം സിലബസ് ടോപ്പിക് പ്രിൻസിപ്പിൾ; പ്രിൻസിപ്പിൾ, ഡയഗ്രാം/പ്രോസീഡർ, അപ്ലിക്കേഷൻ, കോമൺ മിസ്റ്റേക്ക് പ്രിൻസിപ്പിൾ പ്രിൻസിപ്പിൾ.

Prerequisite foundation

Basic electrical engineering, electrical machines, power systems and industrial management.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 • Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 • Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 • Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 • Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain the global and national energy scenario and the provisions of the Energy Conservation Act.
2. Apply energy auditing methodologies to identify energy saving opportunities in various sectors.
3. Analyze energy conservation techniques in electrical systems including motors, transformers and lighting.
4. Evaluate the economic feasibility of energy conservation projects using financial analysis tools.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

Malayalam support: CO ഉള്ളത് exam-ൽ ഉപയോഗിക്കാൻ ഉപയോഗിക്കാൻ ഉപയോഗിക്കാൻ. Define, explain, apply ഉള്ളത് action words ഉപയോഗിക്കാൻ.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain the principles of energy management and the legal framework for energy conservation.	Understand
CO2	Conduct preliminary and detailed energy audits for given industrial or building scenarios.	Apply
CO3	Identify energy conservation opportunities in electrical and thermal utility systems.	Apply
CO4	Perform financial analysis for energy saving measures and prepare energy audit reports.	Apply

Official CO-PO articulation matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7-PO11
CO1	3	2	2	2	2	-	-
CO2	3	2	2	2	2	-	-
CO3	3	2	2	2	2	-	-
CO4	3	2	2	2	2	-	-

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters

Module	Title	Official major topics	Hours	CO	Handbook section
1	Energy Management and Audit Fundamentals	Global and Indian energy scenario; Energy Conservation Act 2001; Energy management principles; Energy audit: definition, objectives, types (Preliminary, Detailed); Energy audit instruments.	Approx. 14 hours	CO1	Module 1
2	Energy Conservation in Electrical Systems	Energy efficient motors (EEM); Variable Frequency Drives (VFD); Transformer loading and efficiency; Power factor improvement; Energy efficient lighting: LED, induction lamps, lighting controls.	Approx. 16 hours	CO2	Module 2
3	Energy Conservation in Thermal Systems	Boiler efficiency and conservation; Steam system optimization; HVAC and Refrigeration energy saving; Waste heat recovery systems; Cogeneration basics.	Approx. 14 hours	CO3	Module 3
4	Energy Economics and Financial Management	Financial analysis of energy saving projects; Simple Payback Period (SPP); Return on Investment (ROI); Net Present Value (NPV); Life Cycle Costing (LCC); Energy Service Companies (ESCOs).	Approx. 16 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Energy Management and Audit Fundamentals

Global and Indian energy scenario; Energy Conservation Act 2001; Energy management principles; Energy audit: definition, objectives, types (Preliminary, Detailed); Energy audit instruments.

CO1 · Approx. 14 hours

Energy Conservation in Electrical Systems

Energy efficient motors (EEM); Variable Frequency Drives (VFD); Transformer loading and efficiency; Power factor improvement; Energy efficient lighting: LED, induction lamps, lighting controls.

CO2 · Approx. 16 hours

Energy Conservation in Thermal Systems

Boiler efficiency and conservation; Steam system optimization; HVAC and Refrigeration energy saving; Waste heat recovery systems; Cogeneration basics.

CO3 · Approx. 14 hours

Energy Economics and Financial Management

Financial analysis of energy saving projects; Simple Payback Period (SPP); Return on Investment (ROI); Net Present Value (NPV); Life Cycle Costing (LCC); Energy Service Companies (ESCOs).

CO4 · Approx. 16 hours

MODULE 1

Energy Management and Audit Fundamentals

Global and Indian energy scenario; Energy Conservation Act 2001; Energy management principles; Energy audit: definition, objectives, types (Preliminary, Detailed); Energy audit instruments.

Approx. 14 hours

CO1

Understand · Apply

Learning goals

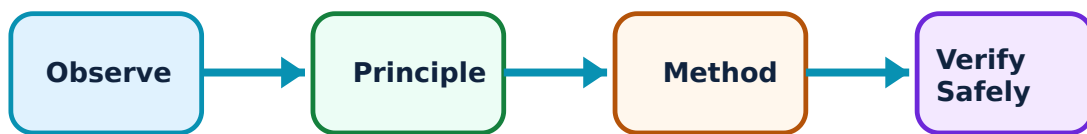
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Energy Management and Audit Fundamentals: identify the system, state its principle, follow the correct method and verify the result safely.

1. Energy Scenario and Legislation–

Energy management involves the strategy of adjusting and optimizing energy, using systems and procedures so as to reduce energy requirements per unit of output. The Energy Conservation Act 2001 provides the legal framework in India, mandating energy audits for designated consumers.

Study and application method

List the main objectives of the 'Energy Conservation Act 2001'; explain the role of a 'Certified Energy Auditor'.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: List the main objectives of the 'Energy Conservation Act 2001'; explain the role of a 'Certified Energy Auditor'.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: ഏതു concept ഈ diagram / circuit / formula / procedure ൽ ഉപയോഗിക്കുന്നു. ഈ result ൽ application ൽ ഉപയോഗിക്കുന്നു. Technical terms English-ൽ എഴുതുക.

Common mistake / precaution: Energy legislation varies by region and sector. Do not provide legal or compliance advice without authorised review of the latest local and national energy regulations.

2. Energy Audit Methodology–

Energy auditing is the key to a systematic approach for decision-making in the area of energy management. A Preliminary Audit uses existing data for quick analysis, while a Detailed Audit involves on-site measurements and in-depth analysis of all energy-using systems.

Study and application method

Compare 'Preliminary' and 'Detailed' energy audits in a conceptual table; list five common instruments used during an energy audit.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare 'Preliminary' and 'Detailed' energy audits in a conceptual table; list five common instruments used during an energy audit.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: ഏതെങ്കിലും concept എഴുതുക. diagram / circuit / formula / procedure എഴുതുക. result application എഴുതുക. Technical terms English-ൽ എഴുതുക.

Common mistake / precaution: Energy auditing involves working with live electrical and thermal systems. Do not perform measurements without authorised safety equipment (PPE) and adherence to industrial safety protocols.

Energy-use calculator

Appliance power (W)

100

Hours used

5

Tariff per unit

6

Enter valid values and choose Calculate.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain

one practical or numerical method without looking at notes.

MODULE 2

Energy Conservation in Electrical Systems

Energy efficient motors (EEM); Variable Frequency Drives (VFD); Transformer loading and efficiency; Power factor improvement; Energy efficient lighting: LED, induction lamps, lighting controls.

Approx. 16 hours

CO2

Understand · Apply

Learning goals

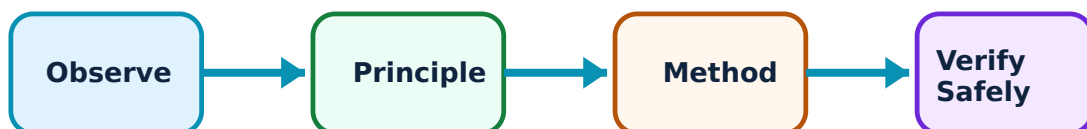
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Energy Conservation in Electrical Systems: identify the system, state its principle, follow the correct method and verify the result safely.

1. Motors and Drives–

Electric motors consume a significant portion of industrial energy. Energy Efficient Motors (EEM) have lower losses due to better materials and design. Variable Frequency Drives (VFD) save energy by matching motor speed to the actual load requirement.

Study and application method

Explain the advantages of 'Energy Efficient Motors' over standard motors; describe how a 'VFD' saves energy in centrifugal pump applications.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain the advantages of 'Energy Efficient Motors' over standard motors; describe how a 'VFD' saves energy in centrifugal pump applications.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: ഏതു concept ഈ problem ഈ diagram / circuit / formula / procedure ഈ result ഈ application ഈ Technical terms English-ൽ ഈ

Common mistake / precaution: Improper VFD installation can cause harmonic distortion. Do not install VFDs without authorised harmonic analysis and proper shielding/grounding.

2. Lighting and Power Factor–

Lighting energy can be reduced by using high-efficiency lamps like LEDs and implementing automatic controls (occupancy sensors, daylight harvesting). Improving power factor reduces

losses in the distribution system and avoids utility penalties.

Study and application method

Calculate the 'Payback Period' conceptually for replacing T12 lamps with LEDs; explain the benefits of 'Capacitor Banks' for power factor correction.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Calculate the 'Payback Period' conceptually for replacing T12 lamps with LEDs; explain the benefits of 'Capacitor Banks' for power factor correction.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: ഏതെങ്കിലും concept നോക്കി അതിന്റെ ഏതെങ്കിലും diagram / circuit / formula / procedure എഴുതുക. ഏതെങ്കിലും result നോക്കി അതിന്റെ application എഴുതുക. Technical terms English-ൽ എഴുതുക.

Common mistake / precaution: Capacitor banks store high-voltage energy even after power-off. Do not perform maintenance on power factor correction units without authorised discharge procedures and safety verification.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

Energy Conservation in Thermal Systems

Boiler efficiency and conservation; Steam system optimization; HVAC and Refrigeration energy saving; Waste heat recovery systems; Cogeneration basics.

Approx. 14 hours

CO3

Understand · Apply

Learning goals

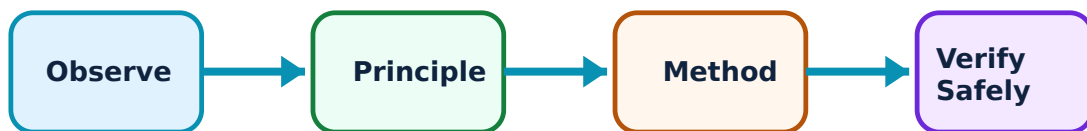
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Energy Conservation in Thermal Systems: identify the system, state its principle, follow the correct method and verify the result safely.

1. Boilers and Steam Systems–

Thermal energy conservation focuses on improving combustion efficiency in boilers and reducing losses in steam distribution. Proper insulation, condensate recovery, and regular maintenance are key to saving fuel.

Study and application method

List four methods to improve 'Boiler Efficiency'; explain the importance of 'Condensate Recovery' in a steam system.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: List four methods to improve 'Boiler Efficiency'; explain the importance of 'Condensate Recovery' in a steam system.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: ഏതു concept ഈ problem ഈ diagram / circuit / formula / procedure ഈ result ഈ application ഈ Technical terms English-ൽ ഈ

Common mistake / precaution: Thermal systems operate at high temperatures and pressures. Do not perform inspections or modifications without authorised pressure-vessel safety clearance and thermal protection.

2. HVAC and Waste Heat Recovery–

HVAC systems are major energy consumers in buildings. Efficiency can be improved through better insulation, high-efficiency chillers, and heat recovery from exhaust air. Waste heat recovery uses heat from industrial processes for other heating needs.

Study and application method

Describe the concept of 'Waste Heat Recovery' with an example; explain how 'Building Automation Systems' (BAS) contribute to energy saving.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe the concept of 'Waste Heat Recovery' with an example; explain how 'Building Automation Systems' (BAS) contribute to energy saving.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: ഏതെങ്കിലും concept നൽകിയിരിക്കുന്ന ചിത്രം / diagram / circuit / formula / procedure നൽകിയിരിക്കുന്നു. ഏതെങ്കിലും result നൽകിയിരിക്കുന്നു application നൽകിയിരിക്കുന്നു. Technical terms English-ൽ നൽകിയിരിക്കുന്നു.

Common mistake / precaution: Refrigerants used in HVAC systems are regulated. Do not handle or charge refrigerants without authorised certification and environmental safety compliance.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Energy Economics and Financial Management

Financial analysis of energy saving projects; Simple Payback Period (SPP); Return on Investment (ROI); Net Present Value (NPV); Life Cycle Costing (LCC); Energy Service Companies (ESCOs).

Approx. 16 hours

CO4

Understand · Apply

Learning goals

Identify core terms, explain the principle and complete the stated application or practical

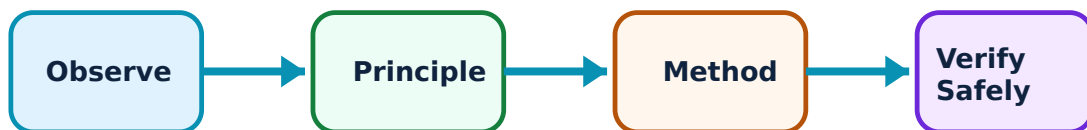
outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Energy Economics and Financial Management: identify the system, state its principle, follow the correct method and verify the result safely.

1. Financial Analysis Tools–

Economic analysis determines the feasibility of energy saving measures. Simple Payback Period (SPP) is the time taken to recover the initial investment. ROI measures the annual return. NPV and LCC provide a more comprehensive view by considering the time value of money.

Study and application method

Calculate the 'Simple Payback Period' for a project with an investment of ₹1,00,000 and annual savings of ₹25,000; define 'Life Cycle Costing'.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Calculate the 'Simple Payback Period' for a project with an investment of ₹1,00,000 and annual savings of ₹25,000; define 'Life Cycle Costing'.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: ഏതു concept നൽകുന്നു എന്ന് പറയുക. ഏതു diagram / circuit / formula / procedure നൽകുന്നു. ഏതു result നൽകുന്നു. application നൽകുന്നു. Technical terms English-ൽ എഴുതുക.

Common mistake / precaution: Financial projections are based on assumptions about future energy prices and interest rates. Do not make investment decisions without authorised financial risk assessment and professional audit.

2. Energy Management Information Systems–

Energy monitoring and targeting involve the systematic collection and analysis of energy data to identify deviations from targets. ESCOs provide energy saving solutions, often through performance-based contracts where they are paid from the savings generated.

Study and application method

Explain the role of an 'Energy Service Company' (ESCO); describe the 'Monitoring and Targeting' (M&T) loop in energy management.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain the role of an 'Energy Service Company' (ESCO); describe the 'Monitoring and Targeting' (M&T) loop in energy management.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: ഏതു concept നൽകിയിരിക്കുന്നു. ഏതു diagram / circuit / formula / procedure ഉപയോഗിക്കണം. ഏതു result ലഭിക്കും application എങ്ങനെ ഉപയോഗിക്കണം. Technical terms English-ൽ എങ്ങനെ എഴുതണം.

Common mistake / precaution: Performance-based contracts involve complex legal and financial terms. Do not enter into ESCO agreements without authorised legal and technical review of the performance guarantees.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

For a comparison: use construction, working, result, advantage and limitation.

Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Energy Management and Audit Fundamentals

Use the concept flow to revise observation, principle, method and verification.

Module 2: Energy Conservation in Electrical Systems

Use the concept flow to revise observation, principle, method and verification.

Module 3: Energy Conservation in Thermal Systems

Use the concept flow to revise observation, principle, method and verification.

Module 4: Energy Economics and Financial Management

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Conduct a preliminary energy audit of your home or classroom and identify three saving opportunities.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Compare LED and Fluorescent lamps in a conceptual table showing energy and cost benefits.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Draw a flowchart of the 'Detailed Energy Audit' process showing the 10-step methodology.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Calculate the annual energy saving and payback period for a given motor replacement scenario.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Identify the 'Star Label' on a household appliance and explain its significance in energy conservation.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure		
Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Energy Management and Audit Fundamentals

Explain Energy Scenario and Legislation and state one correct method, application or precaution. –

Energy management involves the strategy of adjusting and optimizing energy, using systems and procedures so as to reduce energy requirements per unit of output. The Energy Conservation Act 2001 provides the legal framework in India, mandating energy audits for designated consumers.

Answer framework: List the main objectives of the 'Energy Conservation Act 2001'; explain the role of a 'Certified Energy Auditor'.

Important point: Energy legislation varies by region and sector. Do not provide legal or compliance advice without authorised review of the latest local and national energy regulations.

Explain Energy Audit Methodology and state one correct method, application or precaution. –

Energy auditing is the key to a systematic approach for decision-making in the area of energy management. A Preliminary Audit uses existing data for quick analysis, while a Detailed Audit involves on-site measurements and in-depth analysis of all energy-using systems.

Answer framework: Compare 'Preliminary' and 'Detailed' energy audits in a conceptual table; list five common instruments used during an energy audit.

Important point: Energy auditing involves working with live electrical and thermal systems. Do not perform measurements without authorised safety equipment (PPE) and adherence to industrial safety protocols.

Module 2 — Energy Conservation in Electrical Systems

Explain Motors and Drives and state one correct method, application or precaution. –

Electric motors consume a significant portion of industrial energy. Energy Efficient Motors (EEM) have lower losses due to better materials and design. Variable Frequency Drives (VFD) save energy by matching motor speed to the actual load requirement.

Answer framework: Explain the advantages of 'Energy Efficient Motors' over standard motors; describe how a 'VFD' saves energy in centrifugal pump applications.

Important point: Improper VFD installation can cause harmonic distortion. Do not install VFDs without authorised harmonic analysis and proper shielding/grounding.

Explain Lighting and Power Factor and state one correct method, application or precaution.—

Lighting energy can be reduced by using high-efficiency lamps like LEDs and implementing automatic controls (occupancy sensors, daylight harvesting). Improving power factor reduces losses in the distribution system and avoids utility penalties.

Answer framework: Calculate the 'Payback Period' conceptually for replacing T12 lamps with LEDs; explain the benefits of 'Capacitor Banks' for power factor correction.

Important point: Capacitor banks store high-voltage energy even after power-off. Do not perform maintenance on power factor correction units without authorised discharge procedures and safety verification.

Module 3 — Energy Conservation in Thermal Systems

Explain Boilers and Steam Systems and state one correct method, application or precaution.—

Thermal energy conservation focuses on improving combustion efficiency in boilers and reducing losses in steam distribution. Proper insulation, condensate recovery, and regular maintenance are key to saving fuel.

Answer framework: List four methods to improve 'Boiler Efficiency'; explain the importance of 'Condensate Recovery' in a steam system.

Important point: Thermal systems operate at high temperatures and pressures. Do not perform inspections or modifications without authorised pressure-vessel safety clearance and thermal protection.

Explain HVAC and Waste Heat Recovery and state one correct method, application or precaution.—

HVAC systems are major energy consumers in buildings. Efficiency can be improved through better insulation, high-efficiency chillers, and heat recovery from exhaust air. Waste heat recovery uses heat from industrial processes for other heating needs.

Answer framework: Describe the concept of 'Waste Heat Recovery' with an example; explain how 'Building Automation Systems' (BAS) contribute to energy saving.

Important point: Refrigerants used in HVAC systems are regulated. Do not handle or charge refrigerants without authorised certification and environmental safety compliance.

Module 4 — Energy Economics and Financial Management

Explain Financial Analysis Tools and state one correct method, application or precaution.—

Economic analysis determines the feasibility of energy saving measures. Simple Payback Period (SPP) is the time taken to recover the initial investment. ROI measures the annual return. NPV and LCC provide a more comprehensive view by considering the time value of money.

Answer framework: Calculate the 'Simple Payback Period' for a project with an investment of ₹1,00,000 and annual savings of ₹25,000; define 'Life Cycle Costing'.

Important point: Financial projections are based on assumptions about future energy prices and interest rates. Do not make investment decisions without authorised financial risk assessment and professional audit.

Explain Energy Management Information Systems and state one correct method, application or precaution.—

Energy monitoring and targeting involve the systematic collection and analysis of energy data to identify deviations from targets. ESCOs provide energy saving solutions, often through performance-based contracts where they are paid from the savings generated.

Answer framework: Explain the role of an 'Energy Service Company' (ESCO); describe the 'Monitoring and Targeting' (M&T) loop in energy management.

Important point: Performance-based contracts involve complex legal and financial terms. Do not enter into ESCO agreements without authorised legal and technical review of the performance guarantees.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Energy Management and Audit Fundamentals.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Energy Conservation in Electrical Systems.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Energy Conservation in Thermal Systems.

6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Energy Economics and Financial Management.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Global and Indian energy scenario; Energy Conservation Act 2001; Energy management principles; Energy audit: definition, objectives, types (Preliminary, Detailed); Energy audit instruments..
2. Develop a complete answer or practical plan for: Energy efficient motors (EEM); Variable Frequency Drives (VFD); Transformer loading and efficiency; Power factor improvement; Energy efficient lighting: LED, induction lamps, lighting controls..
3. Develop a complete answer or practical plan for: Boiler efficiency and conservation; Steam system optimization; HVAC and Refrigeration energy saving; Waste heat recovery systems; Cogeneration basics..
4. Develop a complete answer or practical plan for: Financial analysis of energy saving projects; Simple Payback Period (SPP); Return on Investment (ROI); Net Present Value (NPV); Life Cycle Costing (LCC); Energy Service Companies (ESCOs)..

LAST-MILE REVISION

Quick Revision

Module 1: Energy Management and Audit Fundamentals

Global and Indian energy scenario; Energy Conservation Act 2001; Energy management principles; Energy audit: definition, objectives, types (Preliminary, Detailed); Energy audit instruments.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Energy Conservation in Electrical Systems

Energy efficient motors (EEM); Variable Frequency Drives (VFD); Transformer loading and efficiency; Power factor improvement; Energy efficient lighting: LED, induction lamps, lighting controls.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Energy Conservation in Thermal Systems

Boiler efficiency and conservation; Steam system optimization; HVAC and Refrigeration energy saving; Waste heat recovery systems; Cogeneration basics.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Energy Economics and Financial Management

Financial analysis of energy saving projects; Simple Payback Period (SPP); Return on Investment (ROI); Net Present Value (NPV); Life Cycle Costing (LCC); Energy Service Companies (ESCOs).

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus

Term / topic	One-line meaning
Energy Scenario and Legislation	Energy management involves the strategy of adjusting and optimizing energy, using systems and procedures so as to reduce energy requirements per unit of output.
Energy Audit Methodology	Energy auditing is the key to a systematic approach for decision-making in the area of energy management.
Motors and Drives	Electric motors consume a significant portion of industrial energy.
Lighting and Power Factor	Lighting energy can be reduced by using high-efficiency lamps like LEDs and implementing automatic controls (occupancy sensors, daylight harvesting).
Boilers and Steam Systems	Thermal energy conservation focuses on improving combustion efficiency in boilers and reducing losses in steam distribution.
HVAC and Waste Heat Recovery	HVAC systems are major energy consumers in buildings.
Financial Analysis Tools	Economic analysis determines the feasibility of energy saving measures.
Energy Management Information Systems	Energy monitoring and targeting involve the systematic collection and analysis of energy data to identify deviations from targets.

LEARNING RESOURCES

References

Recommended energy conservation references

1. BEE (Bureau of Energy Efficiency) Study Material for Energy Auditors.
2. S. C. Tripathy, Utilization of Electrical Energy and Conservation.
3. W. R. Murphy and G. McKay, Energy Management.
4. Paul W. O'Callaghan, Energy Management.

Approved public energy conservation sources

- <https://nptel.ac.in/courses/124105552>
- https://onlinecourses.nptel.ac.in/noc26_ar07/preview

These sources provide the verified study basis while official Course 6033A detail is unavailable; they do not replace official industrial audits, authorised energy management plans, certified equipment specifications or authorised utility plans.